Machine Learning

18-03-26

Vector-Matrix multiplication

$$V_1 = \begin{bmatrix} 1 & 2 \end{bmatrix} \qquad A = \begin{bmatrix} 0 & 1 & 2 \\ 3 & 4 & 5 \end{bmatrix}$$

$$V_1 A = \begin{bmatrix} 1 & 2 \end{bmatrix} \begin{array}{c} \begin{array}{ccc} u_1 & u_2 & u_3 \end{array} \\ \begin{bmatrix} 0 & 1 & 2 \\ 3 & 4 & 5 \end{bmatrix} \end{array}$$

$$V_1 A = \begin{bmatrix} V_1 \cdot u_1 & V_1 \cdot u_2 & V_1 \cdot u_3 \end{bmatrix}$$

$$V_1 A = \begin{bmatrix} 6 & 9 & 12 \end{bmatrix}$$

input $\longrightarrow$ $\boxed{\text{ML model}}$ $\longrightarrow$ output

$V_1 \longrightarrow \boxed{\text{ML model}} \longrightarrow V_1 A$

$V_1 \downarrow \begin{bmatrix} 1 & 2 \end{bmatrix}$

2 input features

$V_1 A \downarrow \begin{bmatrix} 6 & 9 & 12 \end{bmatrix}$

target $\downarrow$ 12

Matrix-Matrix multiplication

$$A = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix} \qquad B = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 0 & 2 \end{bmatrix}$$

2×2 $\qquad$ 2×3

$$a_1 = \begin{bmatrix} 2 & 1 \end{bmatrix} \qquad b_1 = \begin{bmatrix} 1 & 4 \end{bmatrix}$$

$$a_2 = \begin{bmatrix} 0 & 2 \end{bmatrix} \qquad b_2 = \begin{bmatrix} 2 & 0 \end{bmatrix}$$

$$b_3 = \begin{bmatrix} 3 & 2 \end{bmatrix}$$

$$AB = \begin{bmatrix} a_1 b_1 & a_1 b_2 & a_1 b_3 \\ a_2 b_1 & a_2 b_2 & a_2 b_3 \end{bmatrix}$$

$$AB = \begin{bmatrix} 6 & 4 & 9 \\ 8 & 0 & 4 \end{bmatrix}$$

$$\underbrace{\underset{m \times n}{A} \qquad \underset{n \times p}{B}}_{\substack{AB \\ m \times p}}$$

for two matrixes A with $\dim(A) = m \times n$ and B with $\dim(B) = n \times p$ the matrix multiplication is possible only if no. of columns in A is same as number of rows in B.

eg.

house area	rooms	price
100	1	-
120	2	-
150	2	-

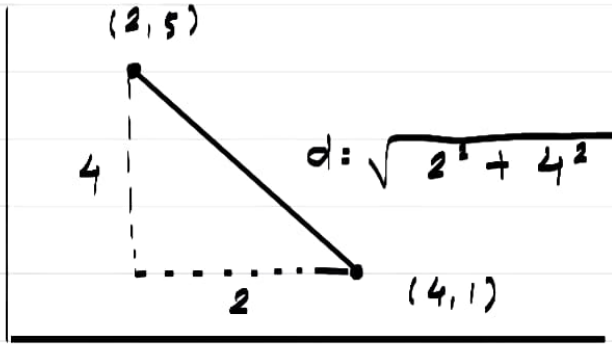
area, rooms $\longrightarrow$ $\boxed{\text{model}}$ $\longrightarrow$ price

input matrix $A = \begin{bmatrix} 100 & 1 \\ 120 & 2 \\ 150 & 2 \end{bmatrix}$ 3×2

model matrix $B = \begin{bmatrix} 1 \\ 10 \end{bmatrix}$ 2×1

$$AB = \begin{bmatrix} 110 \\ 140 \\ 170 \end{bmatrix} \quad \begin{matrix} [100 \quad 1] \\ [120 \quad 2] \\ [150 \quad 2] \end{matrix}$$

Distance between two vectors



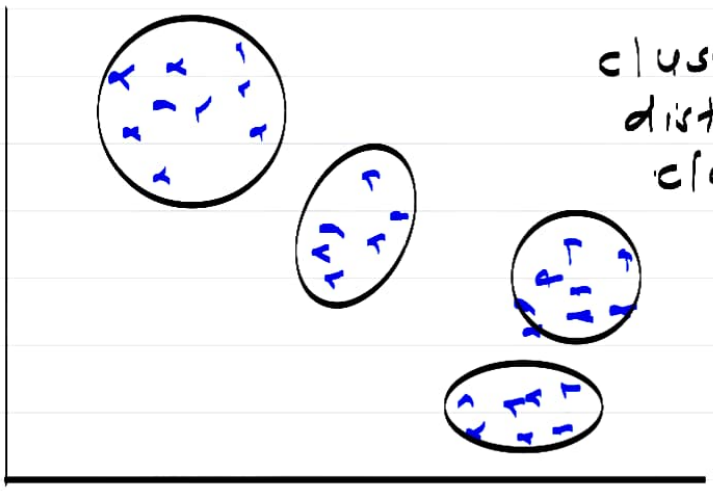
$d = \sqrt{2^2 + 4^2} = \sqrt{20}$ $\longleftarrow$ Euclidean distance

for two vectors

$V_1 = [a_1 \quad a_2 \quad a_3 \ldots a_n]$

$V_2 = [b_1 \quad b_2 \quad b_3 \ldots b_n]$

$$\text{distance}(V_1, V_2) = \sqrt{(a_1-b_1)^2 + (a_2-b_2)^2 \ldots (a_n-b_n)^2}$$



clustering uses euclidean distance to measure closeness of datapoints

Feature Engineering

- its done during data preprocessing step of ML framework

- in FE, we try to find and create features which are important for given ML task

- Less important features are removed from data before model training

this reduces model complexity

model complexity $\propto$ no. of features

it reduces no. of computations, making training faster and stable.

eg. resale car price data

Buy price	age of car	kms driven	Vehicle reg. no	resale price	brand
12	2	1000	1439	5	Tata
13	5	1200	982	7	ms
15	3	1300	312	8	Bmw
⋮	⋮	⋮	⋮	⋮	⋮
✓	✓	✓	✗		✓

this feature is useless to find price of car

we can drop/remove this feature from model training

Methods of finding feature importance

1) Use the domain knowledge

eg. Vehicle reg. no. is not important to find car's price

2) Variance Thresholding

eg. house price pred

area	rooms	city	country	price
1000	2	Hyd	Ind	12
⋮	⋮	Pune	Ind	13
⋮	⋮	Blr	Ind	15
		⋮	⋮	⋮

for all houses, country is fixed as India

As country value not going to change during entire model training, it won't affect the price of house.

- hence we can drop this column

Other column values are varying, they will affect the house price.

country	feature encoded
Ind	1
Ind	1
Ind	1
⋮	⋮
	1

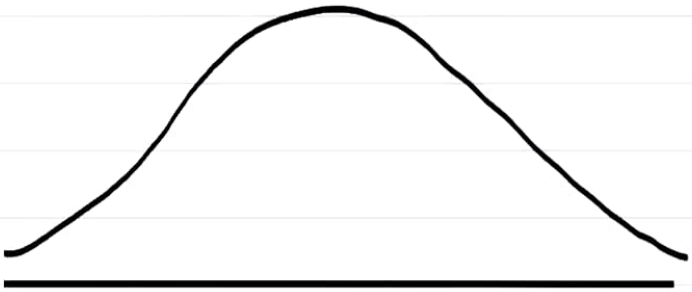
→ mean = 1
variance = 0

as variance is very low for this column, we'll drop it

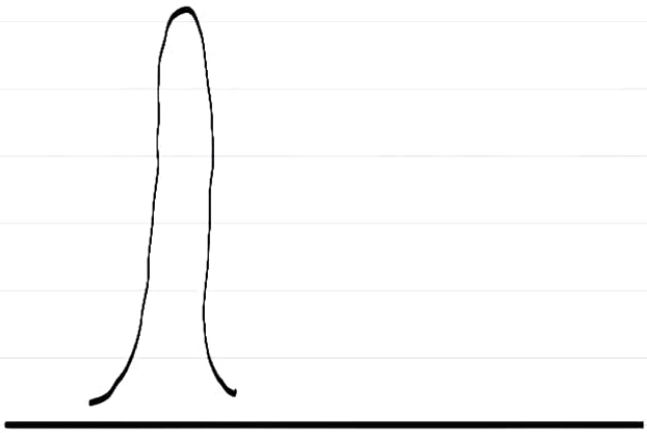
- We'll compute variance of every column/feature

- features having very low or near zero variance will be dropped before training as they don't add any knowledge to the model

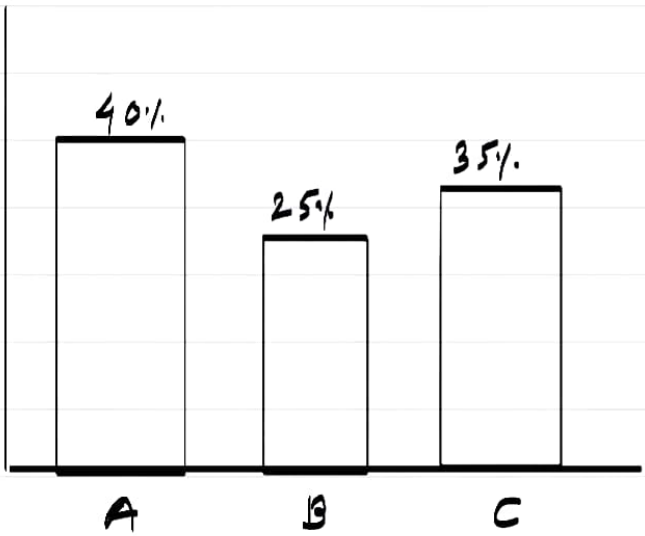
- for categorical features, compute variance after encoding



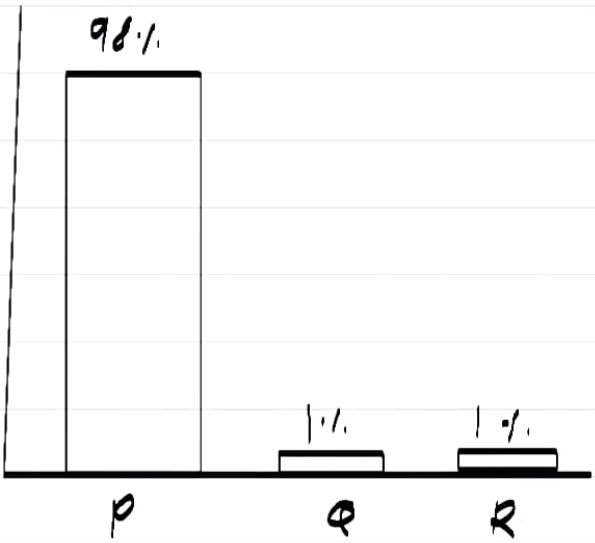
high variance



low variance



high variance



low variance

Feature transformation

eg. land price prediction → rectangular plots

length of plot	breadth of plot	type	price
12	8	Agri	1200
14	7	resident	1400
15	9	⋮	⋮
⋮	⋮	⋮	⋮

- area could be a better feature than length and breadth in this case
- we add this new feature using length and breadth

area (l x b)	type	price
12x8	Agri	⋮
14x9	Res	⋮
15x7	⋮	⋮
⋮	⋮	⋮